

Topic 7: Modern Analytical Techniques I

In order to develop their practical skills, students should be encouraged to carry out a range of practical experiments related to this topic. Hands-on practical work is limited in this topic, although many universities allow students to visit and learn about instrumentation first hand.

Mathematical skills that could be developed in this topic include analysing fragmentation patterns in mass spectra.

Within this topic, students can consider how different instrumental methods can provide evidence for analysis. They can see how accurate and sensitive methods of analysis can be applied to the study of chemical changes, but also to detect drugs such as in blood or urine testing in sport.

Students should:
Topic 7A: Mass spectrometry
1. be able to use data from a mass spectrometer to: - i determine the relative molecular mass of an organic compound from the molecular ion peak - ii suggest possible structures of a simple organic compound from the m/z of the molecular ion and fragmentation patterns